

Fault Injection for ML Applications: From DNNs to LLMs

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I. INTRODUCTION

Machine learning (ML) applications have become ubiquitous in safety-critical and high reliability domains such as transportation, healthcare and finance. Unfortunately, ML applications can fail in unpredictable ways, and hence they should be resilient to faults - this requires performing fault injection on the applications. Fault injection has been an extensively researched and well-explored topic in the DSN community. However, today there is no easy framework that allows ML developers and testers to inject faults, in a customizable fashion. In our group at UBC, we have been developing one such framework called LLTFI over a number of years now (<https://github.com/DependableSystemsLab/LLTFI>). LLTFI is based on and is the successor of another previously developed tool in our group for more than a decade called LLFI (<https://github.com/DependableSystemsLab/LLFI>), which is widely used in academia and industry. LLTFI augments LLFI with the capability to inject faults into ML applications written using a wide-variety of frameworks, such as TensorFlow or PyTorch. We have published a number of papers using LLTFI [1]–[3].

Today, ML applications have moved beyond classical Deep Neural Networks (DNNs) to Large Language Models (LLMs), also known as generative ML. When coming to the era of LLMs, understanding the resilience of LLMs at scale raises new challenges. The error characteristics of faults affecting such models depend on various factors beyond the numerical computation accuracy: unlike traditional HPC workloads that focus strictly on numerical computations, LLM inference is inherently multitasking, simultaneously serving diverse applications such as natural language understanding, sentiment analysis, translation, and interactive conversational tasks. This adaptive behavior complicates resilience characterization because error propagation and the impact of the error become highly dependent on runtime conditions and decisions. A

fault injection methodology that involves above factors would provide a rigorous understanding towards the resilience of LLMs and future AI-driven applications [4], [5].

II. GOALS

This tutorial will teach the basics of using tools to perform fault injection on ML applications, including DNNs and LLMs. We will start by learning to perform simple fault injection experiments for both DNNs and LLMs via LLTFI and a Pytorch-based FI tool. We will also explore how to interpret the results of the fault injection experiments and use them in real-world case studies. Finally, we will delve into large-scale fault injection on LLMs and discuss the techniques, methodologies, and key aspects of FI experiment design.

III. TARGET AUDIENCE

The tutorial targets two groups. The first group is ML developers and testers who use or plan to use fault injection (FI) in their day to day activities. A number of hands-on activities and exercises will be used throughout the tutorial to help this group apply FI to different ML applications and perform trace analysis to understand their resilience under various tasks. The second group are researchers who want to use FI (or similar tools) in their research. The tutorial does not assume any prior knowledge of FI but (some) prior knowledge of ML will be assumed. Though the tutorial will focus on FI of ML applications, the principles discussed can be broadly applicable. We will discuss open issues in the research area of FI for ML applications.

IV. TUTORIAL OUTLINE

- 1) Introduction to FI, Philosophy of FI, and high-level overview of LLTFI (30 minutes)
- 2) Two case studies for illustrating the use of LLTFI, one injecting faults into DNN applications, and the other for injecting faults into LLM applications (60 minutes)

- 3) A case study for illustrating large-scale LLM FI, launching a FI campaign on a cluster, analyzing the obtained FI trace and investigating task-specific LLM outcomes against faults. (60 minutes)
- 4) Open issues and research directions in fault injection, free-form discussion (30 minutes)

V. TAKEAWAYS

The following are the main intended takeaways from the tutorial:

- 1) Understanding of the workflow of performing fault injection on ML applications
- 2) Ability to use LLTFI to inject faults into ML programs (both DNNs and LLMs)
- 3) Ability to perform large-scale fault injection experiment campaigns and analyze the obtained traces
- 4) Knowledge of the open issues and research directions in the area of FI for ML

PRESENTERS

Karthik Pattabiraman is a Professor of Electrical and Computer Engineering (ECE) at the University of British Columbia (UBC) in Canada. He received his PhD and MS in 2009 and 2004 in Computer Science from the University of Illinois at Urbana-Champaign (UIUC). Karthik's research interests are in dependable systems, cyber-physical systems, and software security. For a detailed bio, see: <https://blogs.ubc.ca/karthik/about/full-bio/>

Bo Fang is an assistant professor in the Department of Computer Science and Engineering at the University of Texas at Arlington. He received his Ph.D. in the ECE Department at UBC. His research interests mainly focus on fault-tolerant systems, GPUs and HPC computing in general. He is also working on quantum computing. For a detailed bio see: <https://flyree.github.io/>

Lishan Yang is an assistant professor in the Department of Computer Science at George Mason University. Her research interest mainly falls in system reliability, HPC, and GPU architecture. She received a Ph.D. degree in the Department of Computer Science at William & Mary in 2022. For a detailed bio, see: <https://lishanyang.github.io/>

Abraham Chan is a PhD candidate in Computer Engineering at UBC, advised by Professors Karthik Pattabiraman and Sathish Gopalakrishnan. His primary research interests are in ML reliability, with the goal of developing resilient ML systems against faulty training data in safety-critical applications. For a detailed bio, see <https://people.ece.ubc.ca/abrahamc/>

Yu Sun is a third-year PhD student in the Department of Computer Science at George Mason University, under the supervision of Dr. Lishan Yang. His research focuses on the fault tolerance of LLM applications at scale. For a detailed bio, see <https://pipijing13.github.io/>

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